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Introduction

This report outlines the testing approach for the web application, detailing the tools, methods, and objectives used to ensure it meets functional, non-functional, and accessibility requirements.

Test strategy

Objectives

- To ensure the web-application meets business, functional and non-functional requirements
- Ensure compliance with accessibility standards for users with disabilities.

Scope

- Test user interfaces
- Test database connection
- Test requirements of system
- Test accessibility
- Test security
- Test overall performance of the system

Testing Approach

- **Functional Testing:** Validate user interactions & all specified requirements.
 - Unit Testing (PHPUnit): Test individual units of code to ensure correct functionality.
 - UAT (manual): Validate software meets end-users' requirements & is ready for deployment.
- **Non-Functional Testing:**
 - Performance Testing (Lighthouse): Test for speed, responsiveness, & stability under load.
 - Security Testing (OWASP ZAP): Identify potential vulnerabilities, including input validation, authentication, & session management.
 - Accessibility Testing (WAVE Evaluation): Ensure application is user-friendly & meets user experience expectations.

Unit Testing

PHPUnit

[Inventory-Management-System/tests/Unit at main · JianFGO/Inventory-Management-System](#)

Test Cases

Test Case ID	TC001
Test Description	Verify admin can successfully create a user
Pre-Conditions	1. Admin is logged in 2. Add user form is accessible & properly displayed
Steps to Execute	1. Navigate to Add User page 2. Enter valid full name (e.g. John Doe) 3. Enter valid email (e.g. john@gmail.com) 4. Select role (e.g. Admin) 5. Enter valid password (e.g. password1232) 6. Select branch (e.g. Candy Atlas Sheffield) 7. Click submit button
Expected Result	- User successfully appears on users page - Data saved correctly in database - Success message appears
Status	Pass

Test Case ID	TC002
Test Description	Verify admin can successfully delete a user
Pre-Conditions	1. Admin is logged in 2. Users page is accessible & properly displayed
Steps to Execute	1. Navigate to Users page 2. Select the action 'delete' on the account to be deleted 3. Confirm account deletion
Expected Result	- Account removed from users page - Account removed from database - Success message appears
Status	Fail

Test Case ID	TC003
Test Description	Verify manager can add product
Pre-Conditions	1. Manager is logged in 2. Add Product page is properly displayed
Steps to Execute	1. Navigate to Add Product Page 2. Enter a valid product name (e.g. Pop) 3. Select category (e.g. Lollipops) 4. Select branch (e.g. Candy Atlas Sheffield) depending on manager's branch 5. Enter a valid price (e.g. 0.22) 6. Enter a valid quantity (e.g. 10) 7. Click submit button
Expected Result	- Product successfully created - Data saved correctly in database - Success message appears
Status	Pass

Test Case ID	TC004
Test Description	Verify manager can edit account information
Pre-Conditions	1. Manager is logged in 2. Profile page is accessible & properly displayed with correct details
Steps to Execute	1. Navigate to Your Profile page 2. Edit name (to e.g. Mannie Johnson) 3. Edit email (to e.g. manniejohnson@gmail.com) 4. Click Update Profile
Expected Result	- Profile updated successfully on profile page - Account details updated in database - Success message appears
Status	Pass

Non-Functional Requirements testing

Security

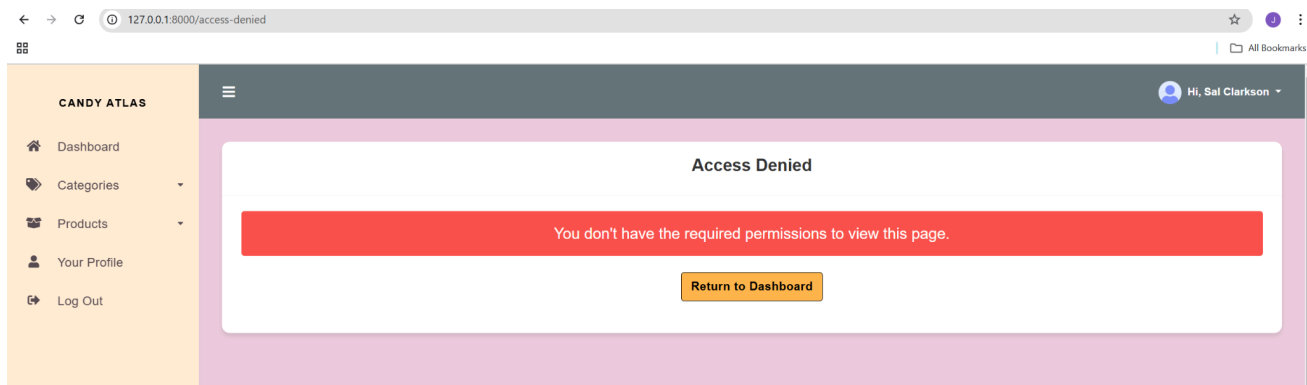
See full scan report at:

<https://github.com/JianFGO/Inventory-Management-System/blob/Main/docs/testing/zap-security-scan-report.pdf>

Steps taken to secure the application:

		id	name	email	email_verified_at	password	remember_token	created_at	updated_at
☐	  		Adel Mint	adel@gmail.com	2025-01-04 22:15:32	\$2y\$12\$M0zAYdTBiO78j0PEpo.MtuBmOsU2H5ATZG1sKRG655...	NULL	2025-01-04 22:15:33	2025-01-04 22:15:33
☐	  	4	Helen Back	helen@gmail.com	2025-01-04 22:15:32	\$2y\$12\$Bpl1CHwRPMZ0y0OWnAoTKuEHYK/MHytL/MivaBKj00...	NULL	2025-01-04 22:15:33	2025-01-04 22:15:33
☐	  	5	Altan Right	altan@gmail.com	2025-01-04 22:15:32	\$2y\$12\$eq/QdU/zzhkOWoPSd6f3qeHPeV4.q4asQ49y0.yRe40...	NULL	2025-01-04 22:15:33	2025-01-04 22:15:33
☐	  	6	Mannie Carter	mannie@gmail.com	2025-01-04 22:15:32	\$2y\$12\$DOJJE0GFMOZ1QTTr01q8ucgDkJIm8XPeWzGgeupPG...	NULL	2025-01-04 22:15:33	2025-01-04 22:15:33
☐	  	7	Sue Pervisor	sue@gmail.com	2025-01-04 22:15:32	\$2y\$12\$SOU4my\$S59Oc/52w6noFJONYWjW1mGHHmM0bRbp5f0...	NULL	2025-01-04 22:15:33	2025-01-04 22:15:33
☐	  	8	Taska Signer	taska@gmail.com	2025-01-04 22:15:33	\$2y\$12\$oodFV6.KEEF5iqhMRdbZWubF2HrNF9UwGN2WiaFvHCp...	NULL	2025-01-04 22:15:33	2025-01-04 22:15:33
☐	  	9	Sal Clarkson	sal@gmail.com	2025-01-04 22:15:33	\$2y\$12\$.taJa5NBbyDjaAx2o/c6v.Mk6lomXgV25SQIDeMNQF/...	NULL	2025-01-04 22:15:33	2025-01-04 22:15:33
☐	  	10	Bill Board	bill@gmail.com	2025-01-04 22:15:33	\$2y\$12\$MXQHIW.lNejc9MjYnJozeu1EOJlxqCmGFnTgYf6clGR...	NULL	2025-01-04 22:15:33	2025-01-04 22:15:33
☐	  	11	Penny Counter	penny@gmail.com	2025-01-04 22:15:33	\$2y\$12\$IGv.v8d7Q5JUDITEMKBzKeO5Er6pgtZ1G0a0bDuQl1Y...	NULL	2025-01-04 22:15:33	2025-01-04 22:15:33

To tackle authentication and authorisation, different tactics have been employed within the program to increase overall security. An example of such is through encryption of account passwords for existing users no matter the role. Authorisation was improved by using the Spatie Laravel package. It is used to manage role-based access control, replacing the roles column in the users table with a dedicated roles table. Blade directives like @role and @endrole were used to adjust page elements based on user roles. An access denied page was added to handle unauthorised access, redirecting users instead of showing a 403 error. For example, if a sales clerk tries to access the orders section, they are redirected to this page. These measures ensure secure and seamless navigation for all users.



New accounts created by admins when logging in for the first time will also get redirected to an email verification page to send a verification email to the account. These methods ensure aspects such as Authentication and Authorisation.

Tool: OWASP ZAP

Browser: Google Chrome

The screenshot shows the OWASP ZAP Manual Explore interface. The top section has a 'Manual Explore' title and a description: 'This screen allows you to launch the browser of your choice so that you can explore your application while proxying through ZAP. The ZAP Heads Up Display (HUD) brings all of the essential ZAP functionality into your browser.' Below this, there are input fields for 'URL to explore:' (http://127.0.0.1:8000/), 'Enable HUD:' (checkbox), and 'Explore your application:' (Launch Browser, Chrome). A note mentions that other browsers can be used but need configuration. The bottom section is a table of HTTP requests and responses.

ID	Source	Req. Timestamp	Method	URL	Code	Reason	RTT	Size Resp. Body	Highest Alert	Note	Tags
18	Proxy	03/01/2025, 11:13:42	GET	http://127.0.0.1:8000/assets/modules/toolip.js	200	OK	21 ms	5,277 bytes			
19	Proxy	03/01/2025, 11:13:42	GET	http://127.0.0.1:8000/assets/modules/popper.js	200	OK	26 ms	19,187 bytes			
20	Proxy	03/01/2025, 11:13:42	GET	http://127.0.0.1:8000/assets/modules/bootstrap.js/boots...	200	OK	5 ms	51,039 bytes	Medium		Comment
22	Proxy	03/01/2025, 11:13:42	GET	http://127.0.0.1:8000/assets/modules/nicescroll/jquery....	200	OK	20 ms	60,010 bytes			
23	Proxy	03/01/2025, 11:13:42	GET	http://127.0.0.1:8000/assets/modules/moment.min.js	200	OK	23 ms	51,599 bytes			
25	Proxy	03/01/2025, 11:13:42	GET	http://127.0.0.1:8000/assets/js/tstla.js	200	OK	18 ms	8,797 bytes			
26	Proxy	03/01/2025, 11:13:42	GET	http://127.0.0.1:8000/assets/js/custom.js	200	OK	15 ms	155 bytes			
27	Proxy	03/01/2025, 11:13:42	GET	http://127.0.0.1:8000/assets/js/scripts.js	200	OK	11 ms	15,948 bytes			
30	Proxy	03/01/2025, 11:13:42	GET	https://content-autofill.googleapis.com/v1/pages/ChVD...	200	OK	126 ms	28 bytes	Low		
34	Proxy	03/01/2025, 11:13:50	POST	https://optimizationguide-pa.googleapis.com/v1/GetMo...	200	OK	142 ms	9,079 bytes	Low		
36	Proxy	03/01/2025, 11:13:51	GET	https://optimizationguide-pa.googleapis.com/download...	200	OK	146 ms	5,195 bytes	Medium		
37	Proxy	03/01/2025, 11:13:51	GET	https://optimizationguide-pa.googleapis.com/download...	200	OK	192 ms	265,059 bytes			
38	Proxy	03/01/2025, 11:13:51	GET	https://optimizationguide-pa.googleapis.com/download...	200	OK	115 ms	883,094 bytes	Medium		Comment
39	Proxy	03/01/2025, 11:13:51	GET	https://optimizationguide-pa.googleapis.com/download...	200	OK	57 ms	45,161 bytes			
40	Proxy	03/01/2025, 11:13:51	GET	https://optimizationguide-pa.googleapis.com/download...	200	OK	347 ms	2,200,227 bytes	Medium		Comment
41	Proxy	03/01/2025, 11:13:51	GET	https://optimizationguide-pa.googleapis.com/download...	200	OK	139 ms	4,680 bytes	Medium		
42	Proxy	03/01/2025, 11:13:51	GET	https://optimizationguide-pa.googleapis.com/download...	200	OK	260 ms	2,747,168 bytes			
43	Proxy	03/01/2025, 11:13:52	GET	https://optimizationguide-pa.googleapis.com/download...	200	OK	125 ms	118,577 bytes	Medium		Comment

Issue 1

The screenshot shows a blue banner with a red flag icon and the text 'Vulnerable JS Library'. Below it, a document icon is next to the text 'GET: http://127.0.0.1:8000/assets/modules/moment.min.js'.

Fix 1

```
PS C:\xampp\htdocs\Inventory-Management-System> npm install moment@latest
>>

added 1 package, and audited 176 packages in 2s

46 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```



momentjs.com/downloads/moment.js

JS moment.min.js

M

JS popper.js

JS sticky-kit.js

Issue 2



Content Security Policy (CSP) Header Not Set (9)

Fix 2



Middleware



ContentSecurityPolicy.php

U

Issue 3



Vulnerable JS Library (2)



GET: http://127.0.0.1:8000/assets/modules/bootstrap/js/boot



GET: http://127.0.0.1:8000/assets/modules/jquery.min.js

Fix 3

JS jquery.min.js

M

Issue 4



Missing Anti-clickjacking Header (2)

Fix 4

```
PS C:\xampp\htdocs\Inventory-Management-System> php artisan make:middleware AntiClickjacking
```

```

use Closure;
use Illuminate\Http\Request;

class AntiClickjacking
{
    /**
     * Handle an incoming request.
     *
     * @param \Illuminate\Http\Request $request

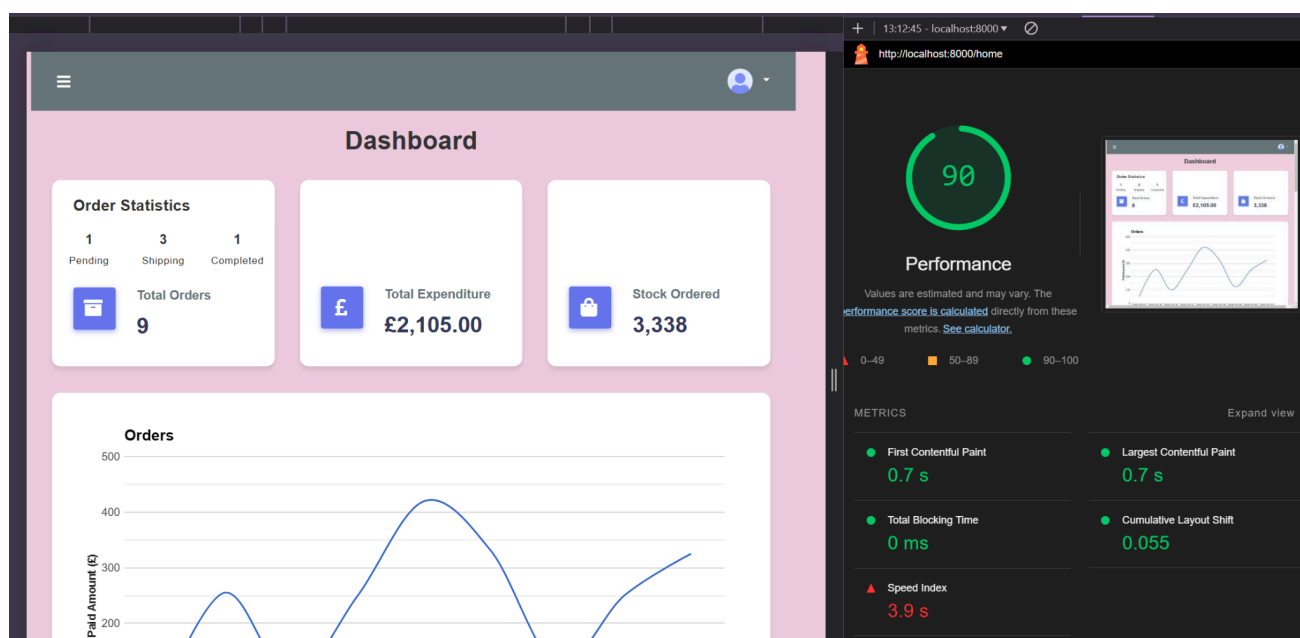
```

Performance

Tool: Lighthouse

Browser: Chrome

Dashboard



Issue 1

▲ Eliminate render-blocking resources — Potential savings of 1,440 ms

Fix 1

```

<!-- General JS Scripts -->
<script src="{{ asset('assets/modules/jquery.min.js') }}" ></script>
<script src="{{ asset('assets/modules/popper.js') }}" defer></script>
<script src="{{ asset('assets/modules/tooltip.js') }}" defer></script>
<script src="{{ asset('assets/modules/bootstrap/js/bootstrap.min.js') }}" defer></script>
<script src="{{ asset('assets/modules/nicescroll/jquery.nicescroll.min.js') }}" defer></script>
<script src="{{ asset('assets/modules/moment.min.js') }}" defer></script>
<script src="{{ asset('assets/js/stisla.js') }}" defer></script>

```

Issue 2

▲ Enable text compression — Potential savings of 990 KiB

Fix 2

```
class GzipMiddleware
{
    /**
     * Handle an incoming request.
     *
     * @param \Closure(\Illuminate\Http\Request): (\Symfony\Component\HttpFoundation\Response) $next
     */
    public function handle(Request $request, Closure $next): Response
    {
        $response = $next($request);
        $response->headers->set('Content-Encoding', 'gzip');
        return $response;
    }
}
```

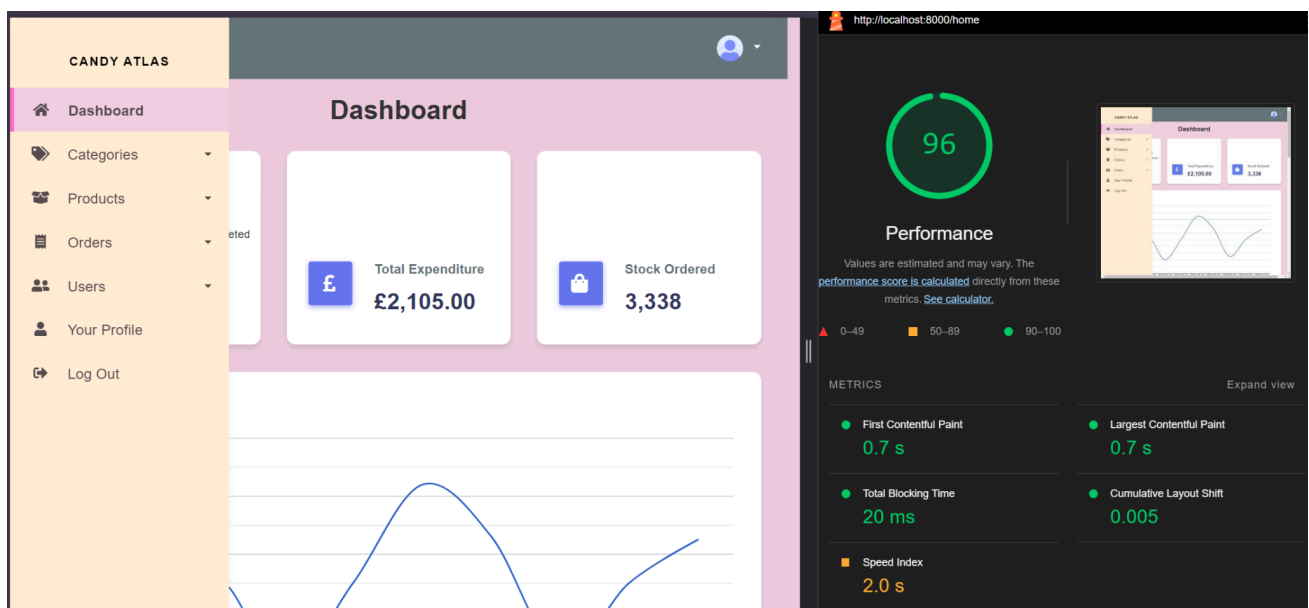
Issue 3

▲ Reduce unused CSS — Potential savings of 304 KiB

Fix 3

```
export default {
    content: [
        './vendor/laravel/framework/src/Illuminate/Pagination/resources/views/*.blade.php',
        './storage/framework/views/*.php',
        './resources/**/*.blade.php',
        './resources/**/*.js',
        './resources/**/*.vue',
        './resources/**/*.scss',
    ],
},
```

Result



Accessibility

Tool: WAVE Evaluation extension

Browser: Microsoft Edge

Login

The screenshot shows the Candy Atlas login page in a Microsoft Edge browser. The WAVE accessibility evaluation tool is overlaid on the left side of the page. The login form includes fields for Email Address (adel@gmail.com) and Password (masked with dots). There is a 'Remember me' checkbox and a 'Login' button. A 'Forgot Your Password?' link is also present. The WAVE tool shows a summary of accessibility issues: 1 Error, 2 Contrast Errors, 3 Alerts, 4 Features, 2 Structural Elements, and 0 ARIA. The Candy Atlas page has a pink background and a white login form.

WAVE powered by WebAIM

Styles: OFF ON

Summary

1 Errors, 2 Contrast Errors, 3 Alerts, 4 Features, 2 Structural Elements, 0 ARIA

View details

h1 Candy Atlas

h4 Login

Email Address: adel@gmail.com

Password: [masked]

Remember me

Login

Forgot Your Password?

No account yet? Real-time admin to create one.

Issue 1

  1 Errors

 1 X Missing or uninformative page title

Fix 1

```
{{-- Browser tab title --}}  
@section('title', 'Login')
```

Issue 2

   2 Contrast Errors

2 X Very low contrast

Fix 2

```
.btn-primary {  
    background-color: #D18706;  
    border-color: #D18706;  
    color: black;  
    font-size: 13px;  
}
```

```
<div class="text-center mt-3">  
    <a style="color: darkblue" href="{ route('password.request') }">Forgot Your Password?</a>  
</div>
```

Issue 3

☒  3 Alerts

☒ 1 X No page regions

☒ 1 X Skipped heading level

☒ 1 X Tabindex

Fix 3

```
<main class="section"> {{-- Previously <section class="section"> --}}
```

```
<div class="card-header">
  {{-- Previously <h4>Login</h4> --}}
  <h2>Login</h2>
</div>
```

```
<button type="submit" style="font-size: 16px" class="btn btn-primary btn-lg btn-block"
  {{-- Removed tabindex="4" --}}>
  Login
</button>
```

Result

The screenshot shows a web browser window with two tabs: "Personal" and "Group 12 ITSE Testing - Google". The address bar shows the URL "127.0.0.1:8000".

Wave Accessibility Evaluation Tool Interface (Left Panel):

- Logo: WAVE powered by WebAIM
- Styles: OFF ☐ ON ☒
- Summary tab selected (other tabs: Details, Reference, Order, Structure, Contrast)
- Summary statistics:
 - Errors: 0
 - Contrast Errors: 0
 - Alerts: 0
 - Features: 4
 - Structural Elements: 3
 - ARIA: 0
- Buttons: View details
- Message: Congratulations! No errors were detected! Manual testing is still necessary to ensure compliance and optimal accessibility.

Candy Atlas Login Page (Main Content Area):

- Header: Candy Atlas
- Section: Login
- Form fields:
 - Email Address: adel@gmail.com
 - Password: [masked]
- Remember me: ☐
- Login button
- Forgot Your Password? link
- Footer: No account yet? Real [Code](#) admin to create one.